Jarvis: Product Requirements Document

Web prototype v0.1 · Working name "Jarvis" · Built for Google AI Studio · All money is simulated

0. Builder instructions (read first)

- Build a responsive web app with React, TypeScript and Tailwind CSS, using client-side routing. Desktop-first, fully usable at mobile widths.
- Persist all state in localStorage behind one storage module, so a backend can replace it later.
- AI features call the Gemini API through a single service module (aiService), using structured JSON output with the schemas in section 8. Never hardcode an API key; use the key mechanism AI Studio provides.
- Every AI call runs live against the Gemini API. There is no mock mode. If a call fails or exceeds 15 seconds, show a clear error message with a Retry button, keep whatever the user typed, and never substitute a fake response.
- Money is simulated. Show a persistent banner: "Prototype: simulated funds, no real money." Do not add payment SDKs.
- Put every tunable number (points, caps, yields, timers) in one config object, exposed in Settings > Advanced.
- Include a **Load demo data** button that seeds 30 days of realistic activity so charts, the scorecard and the wallet look populated.
- Tone: calm, collaborative, never guilt-based. The app is a coach, not a gatekeeper.

1. Overview

Problem. In 2040 every person has a personal AI agent that knows them better than they know themselves. Heavy delegation erodes confidence in one's own judgement: the agent always looks more polished, answers instantly, is free, and is trusted without question.

Concept. Jarvis is an honest, healthy AI workspace that lets people delegate the work but keep the judgement. In the domains a user chooses to grow, it asks them to think first, offers a guided path as well as a direct answer, tracks how their own judgement holds up over time, and rewards independent thinking through a withdrawable pledge and a points-based Judgement Index.

Target user. Time-pressed professionals who delegate heavily to AI. Beachhead: early-career professionals and managers who want to stay credible.

Persona: Arjun, 36, senior manager. Delegates memos, hiring decisions and sensitive messages to his agent. His worst moment is an unexpected question from leadership that he cannot defend in depth. He needs speed where it doesn't matter and sharpness where it does.

North-star metric. Independent Decision Rate (IDR) = decisions in growth domains where the user logged their own attempt before seeing the AI answer ÷ all decisions in growth domains.

2. Product principles (non-negotiable)

1. Delegate the work, keep the judgement. Routine tasks stay instant.
2. The user chooses their growth domains. Effort is never forced elsewhere.
3. Never a dead end. “Show me the answer” is always available.
4. Honest AI. No flattery. Feedback names specific strengths and specific gaps.
5. The deposit is always the user’s: visible, withdrawable at any time, never reduced by penalties.
6. The company never profits from user failure.
7. Sponsors and partners never influence the coach’s hints, resources or rankings.

3. Goals, non-goals and metrics

Goals: raise users’ independent decision rate; improve confidence calibration; keep the app fast enough that users still choose it for real work.

Non-goals (v0.1): real payments, real brand integrations, peer pods, native apps, multi-user accounts.

Metrics

- North star: IDR.
- Inputs: think-first completion rate, hint depth reached, rehearsal completion, teach-back completion, 7-day and 30-day return rate.
- Guardrails: skip rate (Show me the answer without a guess), average time-to-answer, self-reported frustration (thumbs on each session), withdrawal rate.

4. Information architecture

Bottom tabs on mobile, top nav on desktop: **Home · Ask · Growth · Wallet · Settings**. A hidden **Leadership view** route is linked from Settings.

5. Feature specifications

F1. Onboarding and domain setup

- Three steps: (1) “What do you want to stay sharp at?” Pick 1 to 4 **growth domains** from a list (Writing and memos, Decision-making, Hiring and people, Analysis, Presenting, Learning new topics) plus a custom domain. All other areas are **delegation domains** and stay instant.
- (2) Quick baseline: five short questions per growth domain (static question bank, optional AI scoring) to set a starting level (Beginner, Developing, Strong) and a starting Judgement Index of 100.
- (3) Optional pledge (see F12) and plan choice (F15). Both skippable.
- Output: user profile with domains and levels. Deterministic except optional baseline scoring.

F2. Ask flow and mode switch

- Main input: “What do you need help with?” plus a domain selector (auto-suggested by keywords, editable).
- **Mode chip**: *Get it done* or *Help me grow*. In growth domains the default is Help me grow. In delegation domains it is always Get it done and nothing is tracked as a penalty.

- Optional **time budget** chips: 2 min, 5 min, 15 min, No rush. A short budget compresses the guided path (fewer levels) and never blocks the answer.
- Switching a growth-domain question to Get it done is allowed. The first 3 per week are free (the **delegation allowance**); after that it costs points (section 6).

F3. Think-first window

- In Help me grow mode, before anything is revealed, show a focused panel: the question, a text area titled “Your first take”, and a confidence slider (0 to 100%).
- Soft minimum: 20 seconds and 15 words. A gentle timer ring shows elapsed time; it never blocks the user.
- Rotating motivational line beneath the input (see F14). Buttons: **Submit my take** (primary) and **I’m stuck, give me a hint** (leads to F5).
- On submit, call scoreAttempt. If isMeaningful is false (gibberish, filler), ask the user to add a little more; do not award points.
- Output: an Attempt record, then the two-option screen (F4).

F4. Two-option output

After the think-first step, show two cards side by side (stacked on mobile):

- **Classic answer.** The full AI answer from classicAnswer, with its confidence and listed assumptions (see F10). A chip shows the points effect, for example “No change after your take”.
- **Guided path.** The coach’s hint ladder and resources (F5). Chip: “+15 when you complete it”.
- If the user chose Classic without submitting a take, show a short, non-judgmental confirm: “This skips your own thinking in a domain you chose to grow. Continue?” and apply the penalty in section 6.
- Classic is never hidden or delayed beyond that confirm.

F5. Coach: hint ladder and resources

- Five levels, revealed one at a time with a “Next hint” button: (1) a nudge, (2) a guiding question, (3) a framework or checklist, (4) a partial answer, (5) the full answer.
- A progress bar shows levels used. After each level, the user can type or revise their answer; a **Submit final answer** button ends the path.
- **Resources panel** (2 to 4 items): articles, videos or concepts to study, each with a one-line reason. If Google Search grounding is available, return real links. If not, return a searchQuery with url set to null and show it as “Suggested search”. Never invent URLs.
- Difficulty adapts to the user’s level in that domain (from F1 and the scorecard).
- On completion: guided path points, then the compare screen (F6).

F6. Decide-then-compare and calibration

- After the user’s attempt and the AI answer are both available, call compareViews and show a **comparison card** with four lists: Where we agree, Where we differ, What you caught that the AI missed, What the AI caught that you missed. Tone is honest and specific; if the user caught nothing extra, say so plainly.

- The user then chooses **My version**, **AI version** or **Merged**, which is logged as a **Decision** with their stated confidence.
- Outcome capture: after a configurable delay (default 3 days; a **Simulate outcome** button exists for demos), the app asks “How did it turn out?” with Right, Mixed or Wrong. These outcomes feed the **calibration chart**: stated confidence buckets versus actual accuracy.

F7. Defend-it rehearsal

- Entry points: a **Prepare to defend** button on any decision or draft, and a dedicated card on Home (“Have a meeting or review coming up?”).
- The user pastes a draft, decision or memo and selects the audience (Senior leader, Peer, Client, Team).
- The AI plays a sceptical reviewer, asking three questions one at a time (rehearsalQuestions). The user answers in their own words; each answer is scored (scoreRehearsalAnswer) with brief feedback and at most one follow-up.
- Summary: score out of 100, strongest answer, weakest answer, and a short list of topics to revisit that links to the coach. Pass threshold: average score of 60 or more.

F8. Next-day teach-back

- 24 hours after a growth-domain session, a Home card says “Explain yesterday’s decision in your own words”. Three short questions from recallQuestions, answered as free text, scored by scoreRecall.
- Use a **Simulate next day** button for demos. Questions test application and reasoning, not just memory (for example “What would change if the budget were halved?”).

F9. Scorecard, growth map and tracker

- **Judgement scorecard** (Growth tab): IDR over time (line), decisions by source (mine, AI, merged), win rate of the user’s own calls versus AI calls where outcomes are known, calibration chart, rehearsal average.
- **Growth map**: one tile per growth domain showing level, independence percentage, 30-day trend arrow, and last practised date.
- **Tracker**: weekly counts of topics learned, new ideas or drafts started independently, and manual hours (time the user marks as AI-free). Users can add a manual entry. Weekly goal ring per metric (goals editable).
- All deterministic, computed from stored records.

F10. AI mistake and assumptions log

- Every classicAnswer includes confidence, assumptions and “may be wrong if…” items, shown under the answer as **What to verify**.
- Users can tap **Flag an issue** on any AI output (wrong fact, missed context, wrong assumption, tone mismatch) with a short note. Flags are stored in the **AI log**.
- Weekly prompt: “Take two minutes to review where your agent missed this week.” Completing a reflection earns points. The log page shows flag counts by type and domain.
- The log is user-driven. Do not present the model’s self-reported errors as ground truth.

F11. Judgement Index (the stock-style score)

- A single number starting at 100, shown as a line chart on Home and Wallet with 7-day, 30-day and all-time ranges, styled like a calm stock chart (no flashing red or green; use teal for up and a muted amber for down).
- Updated from net points (section 6). Floor of 50. Tapping the chart shows the events behind each move.

F12. Pledge, reward balance and partner coupons (Wallet)

- **Pledge**: the user sets an amount with a slider (₹100 to ₹10,000, default ₹2,000) and can add or withdraw at any time. Withdrawal is instant (simulated), asks for one confirmation, and returns the full principal. Penalties never touch principal.
- **How it works** panel in plain language: "Your deposit stays yours and you can withdraw it anytime. Your pledge is pooled with others and invested in low-risk instruments (simulated). Returns and partner budgets fund your rewards. Going AI-first in your growth domains reduces your reward balance, never your deposit."
- **Reward balance (RC)**: 1 RC equals ₹1 of coupon value. Accrual and reductions follow section 6. Shows balance, pending this week, and lifetime earned.
- **Rewards store**: six sample partner offers, learning-aligned: PageTurn Books ₹100 off (40 RC), LearnLoop 15% off a course (60 RC), Notebook Co. ₹150 off (70 RC), AudioShelf one-month trial (30 RC), FocusDesk Plus one month (50 RC), Marketplace voucher ₹200 (90 RC). Mark them "Sample partner offers". Redeeming deducts RC and shows a mock code.
- **Simulated yield**: assume 6.0% per year on the pledge pool, accrued daily, visible in the Leadership view only. Label it an assumption.
- If principal is withdrawn to zero, reward accrual stops; existing RC remains redeemable.
- Sponsor content is separated from coaching. No partner logos or offers appear inside Ask, Coach or Rehearsal screens.

F13. AI lock

- Settings > AI lock, per growth domain. Two modes: **Idea gate** (the AI answer unlocks after the user writes at least N words of their own take, default 40) and **Time lock** (after a session, the AI is locked for X minutes in that domain, default 30).
- Override: a 10-second countdown plus a one-line reason, costs points (section 6), maximum 3 overrides per day.
- **Urgent** toggle: one tap disables all locks for 60 minutes with no penalty, so the lock never blocks something that truly can't wait.
- Locks apply only to growth domains. Delegation domains are never locked.

F14. Nudges and motivational lines

- Nudge types: streak continues, recall check ready, a rehearsal suggestion, weekly review ready. No guilt wording, no shaming. Maximum 2 per day, with quiet hours in Settings. In-app only for v0.1.

- A library of at least 12 original short lines shown during think-first, for example: “Your first draft is your fingerprint.” “Speed is for tasks. Judgement is for decisions.” “The best question you ask today is your own.” Do not attribute lines to real people.

F15. Plans and feature gating (simulated)

- **Free**: Ask flow with think-first, classic answers, tracker, Judgement Index, pledge and reward balance, 5 guided paths per week.
- **Plus** (placeholder ₹199 per month): unlimited guided paths, rehearsal, teach-back, calibration analytics, full resource coach.
- A plan toggle in Settings simulates upgrading; gated features show a clear upgrade card, never a dead end. No real billing.

6. Points and reward rules

All values live in the config object.

Event	Condition	Points
Think-first attempt	Growth domain, quality score 40 or more	+10 (scaled: 40 to 69 gives 60%, 70 or more gives 100%)
Guided path completed	User submits a final answer	+15
Decision logged (decide-then-compare)	Own take recorded before reveal	+8, plus +4 if final confidence is within 15 points of the outcome
Rehearsal completed	Average score 60 or more	+20 (below 60: +8 for trying)
Teach-back	2 of 3 correct or sound	+10
AI log reflection	Weekly review completed	+8
Manual hour goal met	Weekly goal reached	+15
Classic answer without a take	Growth domain, delegation allowance used up	−6
Lock override	Each override	−5
Classic answer after a take	Any	0 (no bonus, no penalty)

- Daily cap on positive points: +100. Penalties apply only in growth domains.
- **Judgement Index** = max(50, previous + net points × 0.1).
- **Reward balance accrual** each week = min(positive points × ₹0.05, 0.5% of the pledge). Each penalty point reduces reward balance by ₹0.05, with a floor of 0. Principal is never changed by points.

- Anti-gaming: scoreAttempt must judge relevance and substance; copy-pasting the question or filler text scores below 40 and earns nothing.

7. Screens (description)

- **Home**: greeting, Judgement Index mini-chart, today's focus (one suggested action), cards for recall check due and rehearsal, weekly goal rings, quick **Ask** bar.
- **Ask**: question input, domain and mode chips, time budget chips, then the think-first panel, then the two-option screen, then compare.
- **Growth**: scorecard, growth map, tracker, AI log tabs.
- **Wallet**: pledge amount and withdraw/add buttons, reward balance, Judgement Index chart, rewards store, how-it-works panel.
- **Settings**: domains, mode defaults, AI lock, nudges and quiet hours, plan, Advanced config, Load demo data, Leadership view link.

8. AI service specification

Global prompt rules: be honest and specific; never flatter; state uncertainty; keep responses concise; return JSON only, matching the schema; do not invent URLs or sources.

Call	Used in	Input	Output fields
classicAnswer	F4	question, domain	answer, confidencePct, assumptions[], mayBeWrongIf[]
guidedPath	F5	question, domain, userLevel, timeBudget	levels[5] (type, text), resources[] (title, type, why, searchQuery, url or null)
scoreAttempt	F3	question, userAttempt	score 0 to 100, isMeaningful, strengths[], gaps[]
compareViews	F6	question, userAttempt, aiAnswer	agree[], differ[], userCaught[], aiCaught[], suggestedMerge
rehearsalQuestions	F7	draft, audience	questions[3]
scoreRehearsalAnswer	F7	question, answer	score, feedback, followUp or null
recallQuestions	F8	sessionSummary	questions[3]
scoreRecall	F8	question, answer, reference	score, feedback

All AI calls are live; there is no mock mode. Use the Google Search grounding tool for guidedPath resources when available. Deterministic (no AI): timers, points, Judgement Index, wallet and rewards, locks, nudges, tracker, scorecard, plan gating, Leadership view.

9. Data model (localStorage)

- **User**: id, name, plan, config overrides, onboardingDone.

- **Domain**: id, name, isGrowth, level, createdAt.
- **Session**: id, domainId, mode, question, timeBudget, createdAt, usedClassicOnly, hintDepth, helpful rating.
- **Attempt**: id, sessionId, text, confidencePct, score, createdAt.
- **Decision**: id, sessionId, chosen (mine, ai, merged), confidencePct, outcome (right, mixed, wrong, pending), outcomeDueAt.
- **Rehearsal**: id, sessionId or draftText, audience, answers[], scores[], avgScore.
- **Recall**: id, sessionId, dueAt, answers[], scores[].
- **LogEntry** (AI log): id, sessionId, type, note, createdAt.
- **PointEvent**: id, type, points, domainId, createdAt, sessionId.
- **Wallet**: principal, rewardBalance, lifetimeRewards, transactions[] (deposit, withdraw, accrual, penalty, redeem).
- **Coupon**: id, partner, title, cost, redeemedAt, code.
- **Nudge**: id, type, shownAt, dismissedAt.

10. Design and UX

- Calm, modern, spacious. Light and dark themes. Primary accent teal, secondary indigo, warm amber for decreases. Rounded cards, generous whitespace, readable type (16px minimum body).
- Microcopy is collaborative: “Let’s work it out”, “Here’s a hint”, never “I won’t tell you”.
- Accessibility: keyboard navigable, visible focus, sufficient contrast, no information conveyed by colour alone.
- Loading states for every AI call with a short, honest message and a cancel option.
- Empty states always include a next action.

11. Analytics events (local log, viewable in Advanced)

onboarding_completed, ask_submitted, mode_selected, attempt_submitted, classic_chosen, hint_revealed (with level), guided_completed, compare_viewed, decision_logged, outcome_recorded, rehearsal_completed, recall_completed, ai_flagged, lock_override, pledge_changed, withdrawal, coupon_redeemed, plan_changed.

12. Demo script and acceptance criteria

Demo script (Arjun’s day)

1. Load demo data. Home shows an improving Judgement Index and a recall card.
2. Ask a Writing question in Help me grow mode; submit a first take.
3. See the two-option screen; open the guided path, reveal two hints, submit a final answer.
4. View the comparison card; log the decision with a confidence rating.
5. Prepare to defend a draft; answer three sceptical questions; see the score.
6. Open Wallet: change the pledge, see the reward balance move, redeem a sample coupon, withdraw part of the pledge.
7. Turn on an AI lock; trigger an override and see the points effect.
8. Open the Leadership view and adjust assumptions.

Acceptance criteria

- Every AI call shows a loading state and, on failure or after 15 seconds, a clear error with a Retry button. No fake responses are ever shown.
- Withdrawing principal is always possible and never reduced by penalties.
- No penalty ever applies in a delegation domain.
- Classic answers are always reachable within two taps.
- All points, balances and charts reconcile with the PointEvent and Wallet records.
- The simulated-funds banner is visible on every Wallet and Pledge screen.

13. Phase 2 (not in v0.1)

- A one-reviewer feedback flow (share an answer with one trusted person who answers three structured questions) and later full pods with benchmarks.
- Real payments and deposits through a regulated partner, subject to legal review.
- Real partner coupons and employer-funded pledges.
- Native mobile apps and a backend with accounts.
- Optimising flows from user testing and the no-agent task experiment.